

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech Examinations

Assessment – II - (Autumn 2022)

Course Name: Data Structures and Algorithms with C

Course Code: AM607D

Date: 18 – 10 - 2022

Max Marks: 10

Duration: 01 Hr

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams (if any) to be neatly labeled. Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

Q1	Write a program in C to count the number of nodes in a Singly Linked List. <i>Structure function → returns no. of nodes.</i>	[2.5]
Q2 ✓	Design an algorithm/ pseudocode for selection sort. Illustrate the working of selection sort on the following array with 7 elements: 30,45,25,32,55,60,49 <i>(code)</i>	[2.5]
Q3 ✓	Why Binary Search algorithm is more efficient than a linear search? Depict your answer with suitable example? Mention the time complexity level of two algorithms.	[2.5]
Q4	Perform and compute the time complexity of: a) <i>(Explain)</i> <pre> for(int i=n/2;i<=n;i++){ for(int j=1;j<=n;j=j*2){ printf("%d,%d\n",i,j); } } </pre> <i>↓ how</i> b) <pre> int k=1; while(k<=n){ printf("%d\n",k); k=k*2; } </pre>	[2.5]

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ALL THE BEST

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